From Published Evidence to Analyzable Biomedical Cohorts with Scientific Agents

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Abstract

Biomedical studies often report the distributions and associations needed for a new analysis while access to individual records remains limited. We study whether a scientific agent can use this evidence to generate an entire cohort on which researchers can run their intended analyses. The agent identifies what each study measured, translates reported statistics into parameters of a joint model, and records assumptions for relationships the studies leave open. We first tested the workflow where the source model was available. In a longitudinal memory analysis, the agent and a reconstruction of the published generator both recovered a negative interaction below the HbA1c threshold, while their mean estimates for the weaker upper-branch interaction differed in sign. We then evaluated generation from external studies of glycemia and lipids. Six agent constructions produced 18 tables. All six requested quantities agreed in point direction with sample references across the draws, yet the three construction means for residual lipid correlation ranged from 0.108 to 0.141, compared with 0.512 in the reference sample. In a separate evidence comparison, adding published relationships reduced mean absolute discrepancy for five of six regression coefficients. In an earlier prospective glucose study, all 15 agent draws recovered the directions of the BMI and age associations, while eight recovered the sex difference in the BMI slope. Three of five constructions met the joint accuracy criterion, below the four required. Partial evidence can support specific analyses while leaving other relationships unresolved.

CCS Concepts

• **Computing methodologies** → **Modeling and simulation**; • **Applied computing** → *Life and medical sciences*.

Keywords

Synthetic cohorts, biomedical evidence, scientific agents, longitudinal reconstruction

1 Introduction

A scientific question may be ready before the necessary biomedical data are available. A cohort may not have measured a biomarker, individual records from another study may be inaccessible, or no accessible dataset may contain the requested combination of variables. Yet relevant evidence can exist in published means, distributions, adjusted coefficients and subgroup findings. Turning those findings into a complete virtual table would let researchers execute the intended analysis, examine the implications of the evidence and identify what additional information matters. The desired product is scientific data accompanied by an explicit account of how they were constructed.

The difficult step is forming a joint law from heterogeneous evidence. A marginal correlation and an adjusted regression coefficient describe different quantities. Findings also depend on eligibility, measurement, population and adjustment sets. Two correctly extracted coefficients cannot generally be inserted unchanged into one joint equation, and the dependence needed to connect them may be unreported. Consequently, evidence-based generation requires scientific modeling decisions beyond document retrieval and row sampling. Those decisions determine which analyses the resulting dataset can support.

We investigate a workflow in which a scientific agent makes these decisions. The researcher supplies the population, variables, available observations and intended analyses. The agent acquires or receives studies, determines their numerical roles, authors a joint generating law and inspects the resulting data. Numerical tools perform calculations, sampling and analysis; reviews check the statistical meaning and implementation of the specification. The scientific question remains visible to the agent throughout construction. Complete-cohort generation is central: the main experiments supply no individual construction records, and the agent generates every requested variable.

This division of work moves task-specific equations, distribution choices and parameter derivations into the agent's construction process. It also leaves a concrete object to inspect. A researcher can trace a generated association back to a reported statistic, a mathematical derivation or an explicitly assumed relationship. When an assumption changes a scientific conclusion, the workflow can expose the missing evidence instead of hiding it inside a completed table. The approach uses established probability families, allowing the model to change with the request and available evidence.

We formulate cohort construction around a scientific request: the target population, available measurements, admissible studies and proposed analysis. Our method assigns a numerical role to each finding, constructs a joint model, and exposes the assumptions needed to sample complete records. We evaluate the resulting data through the scientific analyses for which they were generated, including adjusted associations, repeated-measures interactions and residual dependence.

We begin with a case in which the generating model is available. In the Health and Retirement Study (HRS) example, the agent and a reconstruction of the published generator receive the same model exports and their simulated cohorts undergo the same longitudinal analysis. The glycemia and lipid studies pose a different problem: they supply evidence from external populations, and the agent must choose how to connect the variables in a joint law. A separate comparison varies which published relationships are supplied. We also report an earlier, prospectively frozen test of the underlying approach. Its target measurements were revealed after all constructions closed. This separates reuse of known references during development from evaluation on a previously withheld sample.

2 Related work and problem definition

2.1 From available information to analyzable records

Filshtein et al. share fitted study models so a recipient can simulate complete cohorts for epidemiological analyses [1]. SynthIPD uses published survival curves and trial summaries to reconstruct time-to-event records with covariates [2]. SynDI combines internal individual data with external regression information across populations to fit a target regression [29]. Multilevel network meta-regression likewise relates aggregate evidence to individual-level models under population assumptions [3]. These approaches differ in what the recipient knows. Filshtein et al. supply the comparison for our HRS experiment because the published exports and the scientific analysis can be matched. In our external-evidence experiments, the agent must also choose the joint dependence structure needed to perform the requested analyses.

G-Sim combines LLM-authored simulators with numerical calibration and structural refinement [4]. LLMSynthor uses aggregate discrepancies to guide proposals for new records and evaluates mobility, e-commerce and household population synthesis [5]. SynthCraft studies healthcare synthetic-data workflows [6], and SPADA samples from a dependency graph fitted with observed tabular data [15]. Our methodological question concerns the preceding biomedical decision: how findings with different populations, scales and adjustment sets constrain one generator, and which analyses the resulting data support.

2.2 Evidence-conditioned cohort construction

A request specifies the target population, eligibility and measurements, desired variables, permitted observations and intended analyses. Each evidence record identifies its source population, measurement scale, statistical operator, adjustment set, estimate, available uncertainty and provenance. The agent returns an executable generator, the numerical uses of that evidence and the assumptions needed to complete the model:

```
(generator, evidence uses, completion assumptions)

    = agent(scientific request, evidence, optional
     observations)

virtual data = generator(number of rows, random seed)

scientific results = requested analyses(virtual data).
```

In whole-cohort construction, optional observations are empty and all requested variables are generated. In enrichment, supplied covariates remain fixed while an unmeasured variable is generated. These modes share the evidence-to-law problem; the principal experiments here concern complete cohorts.

A finding may constrain the law numerically, provide a check on its implications, or remain contextual. Algebraic consequences of accepted findings are derived evidence. Unreported dependence and transfer from a source population to the request are completion assumptions. Several laws can satisfy the available information while implying different conditional results, so the output includes both the table and its construction specification.

We evaluate usefulness at the level of scientific analysis. For an adjusted coefficient, correlation or dispersion statistic, the generated estimate is compared with the relevant sample reference in its native units. Reproducing a supplied summary measures reconstruction; agreement with a target result not supplied to the author addresses a different question. A table can support several analyses with different degrees of evidential support. These distinctions guide both construction and evaluation.

3 Agent-controlled evidence-to-cohort construction

3.1 Qualify evidence against the scientific request

The workflow begins with the population, era, measurements, units, eligibility and scientific analyses. It accepts curated evidence and agent-directed acquisition. With retrieval enabled, the agent selects queries and primary studies to address evidence gaps; the retained record distinguishes full text, partial access and bibliographic metadata and records overlap or exclusions. Researchers supply the request and access conditions, while the agent chooses how the admitted evidence enters the model.

Qualification preserves each finding's original statistical meaning. Logistic disease coefficients cannot become continuous-biomarker slopes, grouped contrasts cannot become per-unit effects, and rank correlations require different treatment from Pearson correlations. Source and requested adjustment sets must also be reconciled. Unit conversion supplies an algebraic transformation rather than evidence of population or assay equivalence. A finding with unresolved definitions can remain useful context without being promoted to a numerical constraint.

In the external-evidence study, researchers supplied audited source packets and measurement guidance [7–11]. Other implementations combined reviewed articles with optional retrieval or allowed authors to select their own sources through a bounded Europe PMC/PMC gateway. The host dispatched authors and coordinated access, review and execution. These implementations share the scientific decision sequence while preserving their separate evidence budgets and input conditions.

3.2 Translate findings into numerical model constraints

The agent authors a model specification, generator code, evidence-use records and a decision explanation. A relationship counts as numerical evidence use when its role is visible in the implemented law or in an explicit calculation. The common evidence-use record connects relationship, dispersion, scenario, check-only and retained-assumption decisions to sources and specification locations. It distinguishes arithmetic from geometric summaries, marginal from conditional effects, rank from Pearson dependence, and individual variation from uncertainty in an estimate.

A worked example shows why this step requires modeling. Jansen et al. report separate age/sex-adjusted HbA1c projections for BMI and glucose [9]. For residualized BMI and glucose with SDs

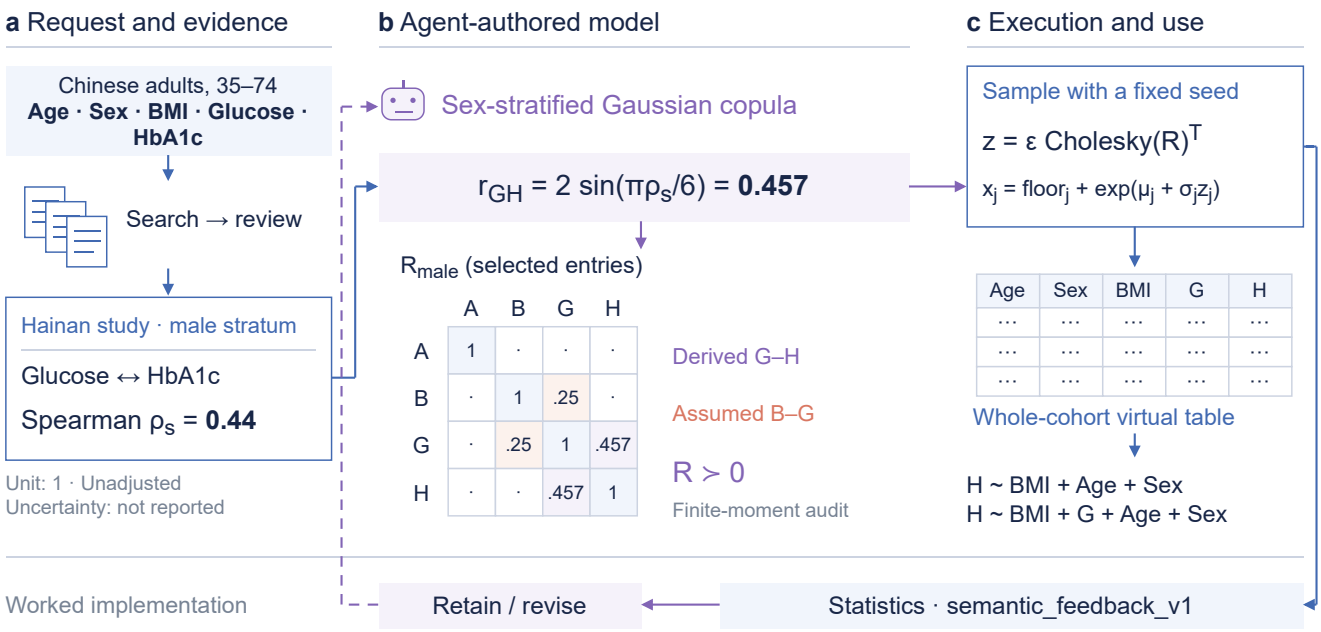


Figure 1: A completed evidence-to-cohort construction. The agent converts the Hainan male glucose–HbA1c Spearman coefficient 0.44 into latent Gaussian correlation 0.457. Selected entries of the sex-stratified copula matrix distinguish the derived glucose–HbA1c relation from the assumed BMI–glucose value 0.25. Standard-normal noise, Cholesky sampling and shifted-lognormal transforms generate BMI, glucose and HbA1c; age has a bounded-integer transform. Table cells are schematic. The scientific law, assumptions, execution checks and requested regressions form one inspectable path. This worked case is separate from the information-contrast and cross-task experiments below.

`sB` and `sG` and correlation `r`, a joint response law with coefficients `thetaB` and `thetaG` implies:

```
cB = thetaB + thetaG * r * sG/sB

cG = thetaG + thetaB * r * sB/sG.
```

CHNS E1 constructor A01, selected by ordinal identifier, used reported slopes `cB = 0.13` and `cG = 2.29`, residual SDs 4.07611 and 0.49694, and the declared assumption `r = 0.25`. Solving the equations gave 0.0642173 and 2.1583154 in IFCC response units. Multiplication by 0.09148 yielded 0.00587460 and 0.19744269 in HbA1c percentage-point units [11]. The source projections therefore informed the joint law through their mathematical relationship, with residual dependence and population transfer remaining explicit assumptions. The worked trace connects the derivation to frozen code, an unfitted correlation check and the resulting target BMI discrepancy.

3.3 Complete an executable joint law

The agent chooses the probability family, dependencies, parameters and remaining completion assumptions. These choices must support complete rows and the requested scientific analyses. Means and SDs alone do not determine residual dependence, tails or subgroup structure, so the specification records how those properties were supplied. Reviews inspect source meaning, numerical derivations, specification/code consistency and required finite moments before execution.

Figure 1 presents a completed construction using a sex-stratified Gaussian copula and positive biomarker margins. The agent converted a reported glucose–HbA1c Spearman relationship into a latent Gaussian correlation, declared the missing BMI–glucose dependence and implemented sampling and measurement transforms. Other requests led to bounded conditional laws, gamma innovations, lognormal mixtures or a recursive longitudinal family. The scientific contribution lies in choosing and connecting these components to the evidence, rather than introducing a new distribution family.

3.4 Execute and inspect the resulting analyses

Deterministic tools execute the authored program through an isolated WSL/bubblewrap adapter and return allowed source and generated-data diagnostics. Semantic checks retain the declared measurement, operator and role; unsupported comparisons remain unavailable. The author records whether to retain or revise the model within the task budget. Candidate versions, final seeds and evaluation contracts are fixed before the corresponding execution or comparison.

Researchers receive the complete tables and their scientific specification, then apply ordinary regression, correlation or longitudinal-analysis software. This separates model authorship from reproducible numerical execution while keeping the analysis request

central to both. Independent numerical implementations check final statistics and reference mappings. The workflow reduces the need for researchers to hand-author each generator equation, although human preparation-time savings were not measured.

4 Evaluation design

Table 1 identifies what each study makes available. HRS supplies fitted model terms, so we can test whether a generated dataset supports the published longitudinal analysis. ENSANUT glycemia and Turkana lipids supply external study evidence, so agreement also depends on the agent's unmeasured dependence and population-transfer assumptions. A construction denotes one authored law. Three final draws from that law describe sampling variation under a fixed model.

CHNS references describe 6,631 selected adults aged 35–74 with complete age, recorded sex, BMI, glucose and HbA1c; analyses are unweighted selected-sample estimates. ENSANUT uses 6,931 selected adults aged 25–79 under fixed eligibility criteria and equal mass across eight sex-by-age cells. The lipid request uses the existing Turkana/Laikipia context: age/sex-adjusted BMI slopes for TC and HDL, plus their residual Pearson correlation after age, sex and BMI adjustment. Frozen unit conversion maps generated lipids to mg/dL. Complete eligibility, analysis definitions and diagnostics are retained in the linked study records and supplement.

Q1 assigns five fresh agent contexts and one prescribed statistical specification to each case/evidence condition, with three final 3,000-row draws per law. Its statistical constructions use the same admitted numerical information and explicit completion assumptions; the asymmetric five-to-one comparison is descriptive. Q5 assigns three fresh contexts to each request. All author sessions close before final reference comparison. Analyses use native units, with seed-averaged absolute discrepancy computed within each Q1 constructor and then averaged across its arm.

In Q1–Q5, CHNS, ENSANUT and Turkana sample references, where used, were previously exposed at project level. In the earlier prospective evaluation, a custodian withheld the exact HAALSI and Turkana reference analyses until all generators and tables were fixed. Investigators had previously seen related Turkana findings. The earlier workflow used source-specific sampling and descriptive checks and preceded the evidence-use interface evaluated in Q5. Q2 deliberately supplies target summaries, and Q3 supplies source-model artifacts. The six cross-task authors share a model service and some sources, while seeds describe sampling under frozen laws. CHNS cluster intervals, ENSANUT cluster-resampling stability bands and lipid HC3 sensitivity intervals describe different reference properties and are not acceptance margins. These conditions permit analysis-specific comparisons, rather than a universal accuracy score.

5 Results

5.1 A shared longitudinal model tests reconstruction

Filshtein et al. distributed fitted HRS model parameters so other researchers could simulate data for longitudinal analysis [1]. We supplied those exports and method code to the agent and separately reconstructed the published recipient-stage generator. Each law produced three complete tables with 11,509 synthetic subjects observed at four visits. We fitted the same mixed model of memory in every table. Its two prespecified interactions estimate how the age slope of standardized memory, per decade, changes per percentage point of HbA1c on each side of the 6.5% spline knot.

Below the knot, the published coefficient was −0.04684. The agent and reconstructed-law estimates averaged −0.04464 and −0.04050 respectively, with all six draws negative (Figure 2). On the upper branch, the published coefficient was +0.01287. All three agent estimates were positive (mean +0.02649) and all three reconstructed-law estimates were negative (mean −0.01169). Five of the six upper-branch model-conditional intervals included zero. Agreement on a slope did not guarantee agreement on individual variation. The agent's upper-group HbA1c SD was 0.563 percentage points, close to the published synthetic summary of 0.6 but lower than the published original-sample SD of 1.4.

The comparison uses six valid tables. Our first three comparator attempts were invalid because our parser misread seven HbA1c coefficient and standard-error pairs. We corrected the parser inputs and reran only the affected comparator draws. The supplement retains all nine attempts. This benchmark measures preservation of supplied model information. The next study asks what can be recovered when the joint model itself must be inferred from the literature.

5.2 External studies constrain some glycemic and lipid analyses

The cross-task study used the common evidence-use interface for HbA1c and TC/HDL requests. All six agent laws completed an initial trial and three final draws, producing 18 distinct 3,000-row tables, or 54,000 fictional rows. H01–H03 chose different bounded conditional laws and HbA1c innovations. L01–L03 combined Gaussian copulas and lognormal margins with differing uses of geometric summaries, arithmetic moments, coefficients of variation and age dependence [25–28]. These choices were authored by the agents, with their derivations and assumptions recorded before execution.

All 54 primary point estimates agreed in direction with their corresponding cached sample estimates (Figure 3; Table S5), while magnitudes varied. HbA1c age effects averaged 0.09954–0.10091 percentage points per decade versus 0.07513 in the reference. Age/sex-adjusted BMI effects averaged 0.04892–0.06030 versus 0.08364 per 5 kg/m². After adding glucose, H03 averaged 0.05098 versus 0.05664, whereas H01/H02 averaged 0.03047/0.01936. The tables therefore supported the intended regressions and retained the attenuation pattern after glucose adjustment, with unequal quantitative agreement.

Lipid results made the distinction between separate effects and joint dependence especially clear. HDL BMI effects averaged −2.1865 to −1.9868 mg/dL per 5 kg/m² versus −4.6230; TC effects averaged 10.0151–11.2609 versus 5.6831. The TC reference sensitivity interval crossed zero, so its positive sample point is not an established positive population direction. Across constructors, mean adjusted residual TC–HDL correlations ranged from 0.10828 to 0.14134 versus 0.51207. The laws reproduced focal point directions while substantially underrepresenting this remaining joint association.

Table 1: Information conditions and scientific evaluation. All principal constructions generate complete cohorts without individual-level construction inputs. Historical study identities Q1–Q5 are retained across the main paper and supplement.

Study	Information available to the constructor	Constructions / final data	Scientific assessment
Q3: longitudinal reconstruction	Published model artifacts and a mixed-model analysis	1 agent + 1 reconstructed law; 6 valid tables of 11,509 subjects × 4 visits	Published longitudinal effects and subgroup spread
Q5: glycemia and lipid requests	Reviewed external articles, population metadata and optional retrieval	6 agent laws; 18 × 3,000 rows	HbA1c/TC/HDL effects, residual dependence and dispersion
Earlier prospective evaluation	External summaries and target descriptions, before disclosure of target estimates	5 agent laws + 1 prescribed law per case, 36 target tables of 5,000 rows in total	HAALSI glucose associations and sex interaction, Turkana lipid analyses
Q1: relational-evidence contrast	Basic/marginal evidence E0; additional relational evidence E1	20 agent + 4 statistical laws; 72 × 3,000 rows	Six CHNS/ENSANUT regression coefficients and complete diagnostics
Q2: summary reconstruction	Literature L; literature plus 30 target summaries R	6 agent laws; 18 distinct final tables × 3,000 rows	Joint reconstruction of supplied regression and correlation summaries
Q4: open retrieval	Requests and bounded retrieval; no bibliographic seeds	4 agent laws; 12 × 3,000 rows	Repeated CHNS/ENSANUT construction and scientific analyses
Supporting dependence example	Five retrieved studies and an analysis request	1 agent law; 3 × 3,000 rows	Complete effects and a pre-generation analytical sensitivity

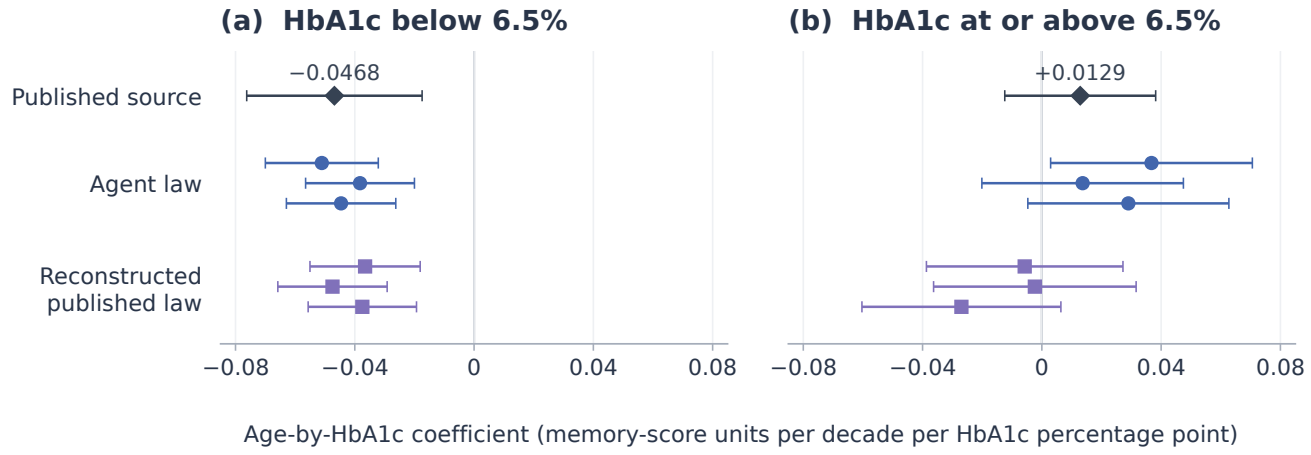


Figure 2: Two prespecified age–HbA1c interactions in a longitudinal memory model. Diamonds show the supplied coefficients; circles and squares show estimates from three draws per law (seeds 6101–6103, top to bottom). Bars show approximate 95% Wald intervals calculated from the exported source standard errors or from the generated-table fits. Units are standardized memory-score change per decade per percentage point of HbA1c within the indicated branch. Each table contains 11,509 synthetic subjects and four visits per subject. The upper-branch signs differed between constructions. The six tables describe sampling under two frozen generating laws.

Dispersion further distinguished the analyses these tables might support. H03's relatively close glucose-adjusted BMI effect coexisted with HbA1c marginal SD/residual RMS of 0.3523/0.3006 versus 0.5274/0.4885. H02 retained more spread but smaller BMI effects. L02's TC SD was close to the reference, 43.6778 versus 42.8466 mg/dL, while its residual lipid correlation remained weak. No law uniformly dominated focal effects, dependence and dispersion. Independent recomputation verified all 513 prespecified estimates and diagnostics, including 324 matched cached-reference comparisons; 189 diagnostics lacked matching references and remained generated-only results.

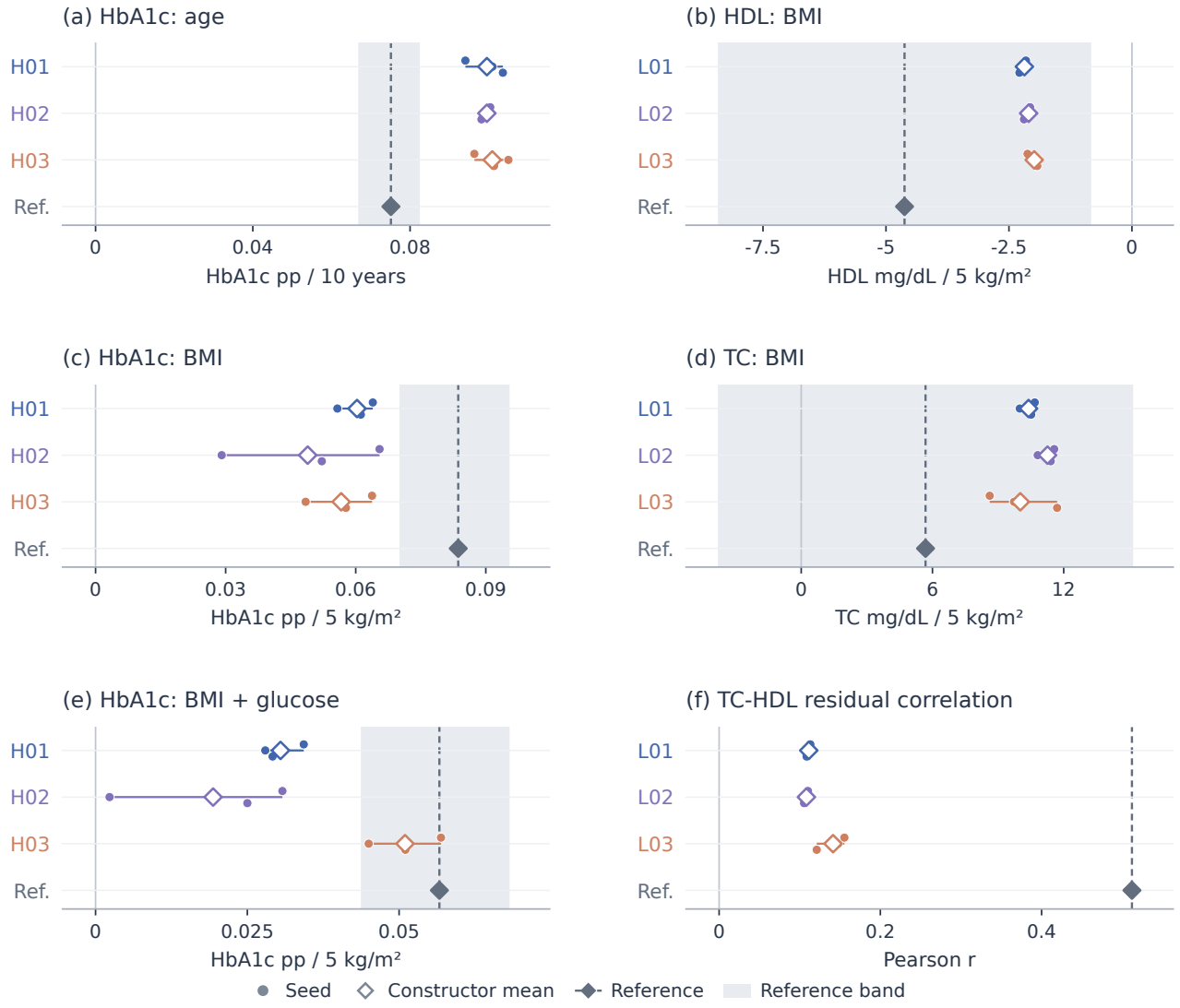


Figure 3: Analysis-specific behavior across glycemia and lipid requests. HbA1c panels in the left column show age adjusted for sex/BMI/glucose (a), BMI adjusted for age/sex (c), and BMI additionally adjusted for glucose (e). The right column shows age/sex-adjusted HDL and TC BMI slopes (b,d) and residual Pearson correlation after age/sex/BMI adjustment (f). All six laws and 54 estimates from 18 tables are retained. Circles show seeds, open diamonds means and thin segments seed ranges; dark diamonds and dashed lines mark sample references. HbA1c units are percentage points per 10 years or 5 kg/m²; lipid slopes are mg/dL per 5 kg/m². Bands are ENSANUT empirical cluster-resampling stability bands and lipid HC3 assumption-conditional sensitivity intervals. The TC band crosses zero; no residual-correlation interval is available. Seed ranges describe sampling under each frozen law. Table S5 supplies all numerical summaries and Figure S3 the dispersion results.

5.3 Scientific use before disclosure of the measured reference

The earlier prospective study defined three analyses of capillary glucose in HAALSI adults aged 40 or older [30,31]. All five agent constructions, each sampled three times, recovered the positive directions of the adjusted BMI and age effects in the 4,394-person sample. The measured difference between female and male BMI slopes was −0.244 mmol/L per 5 kg/m². Only eight of 15 draws recovered its negative direction (Table S7, Figure S5).

The prespecified rule required four of five constructions to satisfy the margin for every analysis on every seed. Three passed in HAALSI and none in the companion Turkana evaluation [32]. In HAALSI, setting all three effects to zero also satisfied the 0.10 outcome-SD margin. This margin alone could not establish recovery

of a nonzero relationship. The planned expansion stopped. The later Turkana analysis in Section 5.2 reused the same measured sample.

5.4 Relational evidence improves analysis agreement

All 24 Q1 constructions produced their three final tables, yielding 72 distinct tables and all 216 focal coefficient estimates. Additional relational evidence reduced constructor-averaged mean absolute discrepancy for five of six analyses (Figure 4; Table S1). For CHNS glucose, discrepancy fell from 0.134019 to 0.024079 HbA1c percentage points per mmol/L. All three ENSANUT contrasts improved. CHNS BMI instead worsened, from 0.020380 to 0.031557, showing that additional information did not improve every requested effect.

The individual laws explain the aggregate contrast. CHNS E1 glucose means spanned 0.363363–0.414189 around the 0.402093 reference; BMI means were positive but lower than the reference. Conditional-age means ranged from −0.000509 to 0.007393 versus 0.008107, including one negative constructor mean. Every E1 author used at least one relation numerically, yet useful source information did not uniquely determine the target joint law. The prescribed E1 statistical constructions also had lower discrepancies than every corresponding agent for CHNS glucose and ENSANUT conditional BMI; other comparisons favored agent laws. The result supports the value of relational information and inspectable model construction without an overall algorithmic ranking.

5.5 Supplied summaries enable joint reconstruction

The summary-assisted study tests a different information condition: three authors received literature alone (L), while three received that literature plus 30 target summaries and their metadata (R). Each delivered three final tables. R closely reconstructed all five highlighted regression and correlation analyses together (Figure S1; Table S2). Constructor-mean BMI coefficients ranged from 0.052821 to 0.055789 around the supplied 0.054946; glucose coefficients ranged from 0.396941 to 0.413715 around 0.402093. Both Pearson and Spearman glucose–HbA1c correlations were also close, preserving joint features that differed in several external-evidence constructions.

Agreement extended to spread: R constructor-mean HbA1c SDs were 0.95208–0.96554 versus 0.95357. A separate purpose-built statistical calibration supplement likewise closely reconstructed the regression summaries. These results show that available aggregate information can become useful joint data. They measure reconstruction of supplied information and remain separate from Q1's external-evidence comparison.

5.6 Retrieval and sensitivity to an unreported relationship

Four open-retrieval authors selected sixteen paper uses representing twelve originals and produced twelve complete CHNS/ENSANUT tables (Figure S2; Tables S3–S4). All 36 primary points retained their reference directions, while rank and residual-spread discrepancies remained.

The supporting dependence example illustrates another useful outcome of explicit construction. Before generation, two admissible BMI–glucose latent correlations, 0.10 and 0.40, implied conditional BMI coefficients +0.03422 and −0.01095 while preserving margins and the secondary projection (Figure S4). Making the law explicit identified an unreported relationship capable of changing the requested conditional conclusion.

6 Discussion

Scientific requests and study-qualified partial evidence can become executable joint laws and analyzable virtual cohorts without individual-level construction records. The agent's central role is choosing how evidence enters that law: qualifying study findings, deriving compatible numerical constraints and completing unreported relationships. The table makes those choices usable by ordinary scientific software; the specification makes their consequences inspectable. Established distributions and deterministic numerical tools provide the implementation, while the agent supplies the task-specific scientific model.

The HRS reconstruction shows what happens when a published study supplies the model terms. Both constructions preserved the lower-branch interaction, yet the weaker upper-branch point estimates differed in sign. Under external evidence, a similarly interpretable distinction emerged: an ENSANUT generator could approximate the glucose-adjusted BMI coefficient while understating HbA1c variation, and a Turkana generator could approximate total cholesterol dispersion while understating residual dependence. Each comparison evaluates a specific use of the generated data. More synthetic rows would narrow Monte Carlo variation under the chosen law while leaving the omitted dependence unresolved.

For the present cross-task outputs, supported use is assumption-dependent scenario data for qualitative association exploration and analytical workflow inspection. Quantitative cohort replacement, calibrated design/power use and broad population transfer remain unestablished. These are limits of the present evidence, while the general method continues to address whole-cohort generation, enrichment and eventual missing-cell completion. The latter remains unevaluated. The earlier HAALSI evaluation established prospective direction agreement for two main effects, with mixed results for their interaction and an unsuccessful joint decision. Assessing transport of the later evidence-use interface would require a separate, previously unexposed target.

The current studies also leave component-level questions open. All six cross-task authors retained their initial laws, and all 132 runtime source-check entries were unavailable under the semantic rules (Table S6). Numerical evidence still entered reviewed parameter derivations, but this batch demonstrates no numerical source-fit feedback or revision gain. Likewise, the studies do not isolate a benefit of retrieval, review or a particular probability family. Fresh contexts share model services and some sources, and neither seeded draws nor conditional intervals quantify evidence, extraction or structural uncertainty. Human time savings and arbitrary-request autonomy require separate assessment. The strongest current result is a completed, inspectable scientific workflow with observable analytical successes and failures.

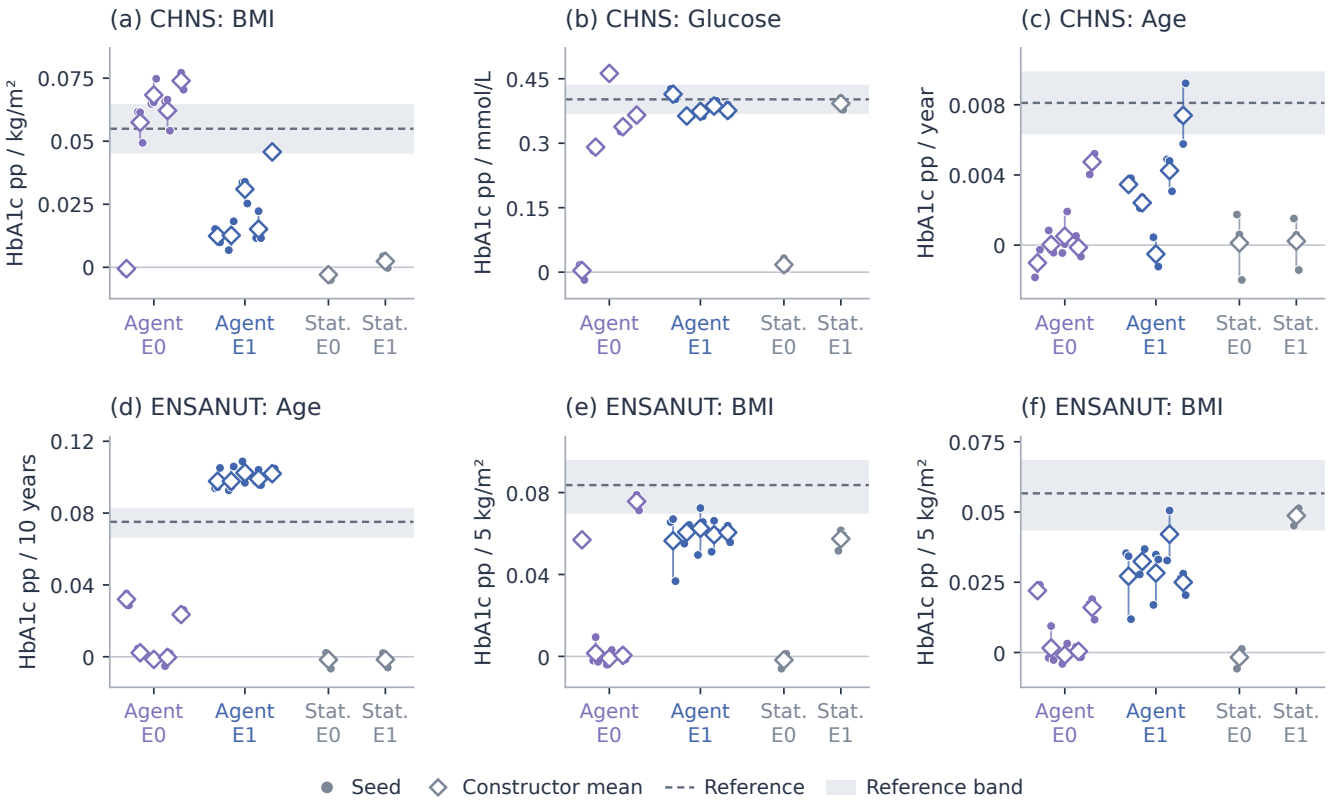


Figure 4: Relational evidence and scientific analysis agreement. CHNS panels show BMI adjusted for age/sex (a), glucose adjusted for age/sex (b), and age adjusted for sex/BMI/glucose (c). ENSANUT panels show age adjusted for sex/BMI/glucose (d), BMI adjusted for age/sex (e), and BMI additionally adjusted for glucose (f). All six analyses retain native units. E0 contains basic/marginal evidence; E1 adds relations. Each agent arm has five independently authored laws and each statistical arm one prescribed law, with all three final draws retained: 24 constructions, 72 tables and 216 coefficient estimates. Dots show seeds, open diamonds constructor means and thin segments seed ranges; dashed lines mark sample references. Reference bands are CHNS community-cluster CR1-t intervals or ENSANUT cluster-resampling stability bands. Author indices do not pair E0 with E1. Table S1 gives every endpoint's mean absolute discrepancy and E1–E0 contrast.

Supporting materials and reproducibility

The supplementary results preserve the study designs, estimators, source inventories, implementation incidents and numerical audits.

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Supplementary results

The supplement retains the complete numerical comparisons and supporting displays under their original study identities. Figure and table numbers follow their new placement; no construction, primary result or unavailable comparison is removed by the shorter main narrative. Q1–Q5 remain distinct experiments with their original information conditions.

S1. Complete external-evidence contrast

All 24 Q1 constructions produced their three scheduled final tables. The 72 final tables were distinct and supported every requested regression and diagnostic; seven exploratory tables brought physical executions to 79. Additional relational evidence reduced constructor-averaged mean absolute discrepancy for five of the six designated coefficients (Table S1). CHNS glucose discrepancy decreased from 0.134019 to 0.024079 HbA1c percentage points per mmol/L. All three ENSANUT contrasts improved. CHNS BMI discrepancy increased from 0.020380 to 0.031557.

The effect displays explain what these changes mean. CHNS E1 glucose constructor means were 0.363363–0.414189 around the 0.402093 reference. Its BMI means were positive but lower, 0.012400–0.045757 versus 0.054946. Conditional-age means ranged from −0.000509 to 0.007393 versus 0.008107; one author retained a negative mean and two negative seeds. All ENSANUT E1 focal coefficients were positive, although its age means of 0.097769–0.102452 per decade exceeded 0.075132 and its BMI means remained below their references. Preserving a direction and recovering its magnitude are distinct aspects of usefulness.

Every E1 author used at least one relationship as a numerical constraint. The complete uptake records include constraints, unfitted analytic checks and contextual uses, including the projection calculation in Section 3.2. Statistical references also achieved useful results: under E1, the prescribed constructions had lower seed-averaged discrepancies than every corresponding agent for CHNS glucose and ENSANUT conditional BMI. Other comparisons favored agent constructions. The evidence therefore supports scientific utility and the value of additional information, rather than a universal algorithmic ranking.

S2. Complete supplied-summary reconstruction

The separate CHNS experiment gave three authors external literature (L) and three authors that literature plus 30 target summaries and their metadata (R). Each author delivered three tables and had one additional diagnostic-review opportunity. All six retained their models. There were 42 physical executions: six trials and 36 initial/final deliveries. Final bytes duplicated their matching initial outputs, leaving 24 distinct tables overall and 18 distinct delivery tables. The identical stages are displayed once.

R supported close reconstruction of all five highlighted scientific analyses (Table S2; Figure S1). These values were legitimate construction inputs; their agreement measures reconstruction rather than prediction of hidden coefficients. The L/R packet and authors differ from Q1, so those two information comparisons are kept separate.

Agreement extended beyond central values. R constructor-mean BMI SDs were 3.38965–3.40243 versus 3.40632, glucose SDs were 1.48817–1.58968 versus 1.54455, and HbA1c SDs were 0.95208–0.96554 versus 0.95357. A separate three-table statistical calibration supplement also closely reconstructed the regression summaries, with mean coefficients 0.055726, 0.412568 and 0.007679. That purpose-built family was developed with coding-agent assistance and is not an agent-removal ablation. Both approaches demonstrated the usefulness of constructing a joint table from accessible aggregate information.

S3. Open retrieval across population requests

The unified workflow allocated two fresh scientific authors to each of the existing CHNS and ENSANUT requests. Each author received the intended analyses, population/measurement definitions, task interface and retrieval budget, but no target coefficients, desired signs, individual records or prior construction. CHNS- and ENSANUT-derived studies were excluded as generating evidence. Across 24 query credits, the authors selected sixteen paper uses representing twelve distinct originals. Independent review checked the original statistical definitions, numerical derivations, source applicability and required finite moments before execution. Sources could overlap across authors; four constructions do not imply four independent bodies of clinical evidence.

The agents chose materially different laws. C01 used an age-conditioned multivariate-lognormal mixture informed by Chinese population summaries [7,16] and contextual age/body-size findings [10,17]. C02 combined copula predictors with an affine HbA1c conditional mean using Pinggu summaries and Singapore coefficients [18,19], with other populations as applicability context [14,20]. E01 used bounded beta predictors and a conditional gamma response [10,17,21,22]. E02 combined lognormal BMI, conditional truncated-normal glucose and a positive conditional-mean lognormal response [10,21,23,24]. Their unreported dependencies, population transfers and residual choices remained declared assumptions. The shared controller did not prescribe these families or insert target coefficients.

Each author completed one 3,000-row trial, inspected generated-only diagnostics and source feedback, then selected its final model before three fixed draws. All four retained the initial law. The sixteen physical executions comprised four trials and twelve unique final tables, or 36,000 final virtual rows. Feedback contained 29 qualified approximate comparisons and 39 unavailable checks across the four trials, not source-population replays or 29 passed constraints. After all sessions closed, the evaluator reused the fixed sample references. A separate numerical implementation reproduced all 366 final coefficient/diagnostic values; no clinical data were downloaded or refitted.

All 36 primary seed estimates had the same positive directions as their corresponding references (Figure S2). These are repeated measurements of six analysis definitions across four laws, not 36 independent confirmations. Magnitude recovery varied by analysis. CHNS C01/C02 means were 0.03992/0.05338 for BMI against 0.05495, 0.37754/0.49241 for glucose against 0.40209, and 0.00311/0.00563 for age against 0.00811. Units are respectively HbA1c percentage points per kg/m², mmol/L and year. C02 was closer for BMI and age, while C01 was closer for glucose; no construction dominated all analyses.

Table S1: Complete main-study information contrast. Discrepancies retain the units of each row; they are not pooled. The reference response is HbA1c percentage points. E1–E0 uses unrounded quantities before display rounding.

Case / focal analysis	Predictor increment	E0 mean absolute discrepancy	E1 mean absolute discrepancy	E1 − E0
CHNS BMI, adjusted age/sex	1 kg/m²	0.020380	0.031557	+0.011178
CHNS glucose, adjusted age/sex	1 mmol/L	0.134019	0.024079	−0.109939
CHNS age, adjusted sex/BMI/glucose	1 year	0.007281	0.004854	−0.002426
ENSANUT age, adjusted sex/BMI/glucose	10 years	0.063944	0.024670	−0.039274
ENSANUT BMI, adjusted age/sex	5 kg/m²	0.056883	0.023732	−0.033151
ENSANUT BMI, adjusted age/sex/glucose	5 kg/m²	0.048797	0.025649	−0.023147

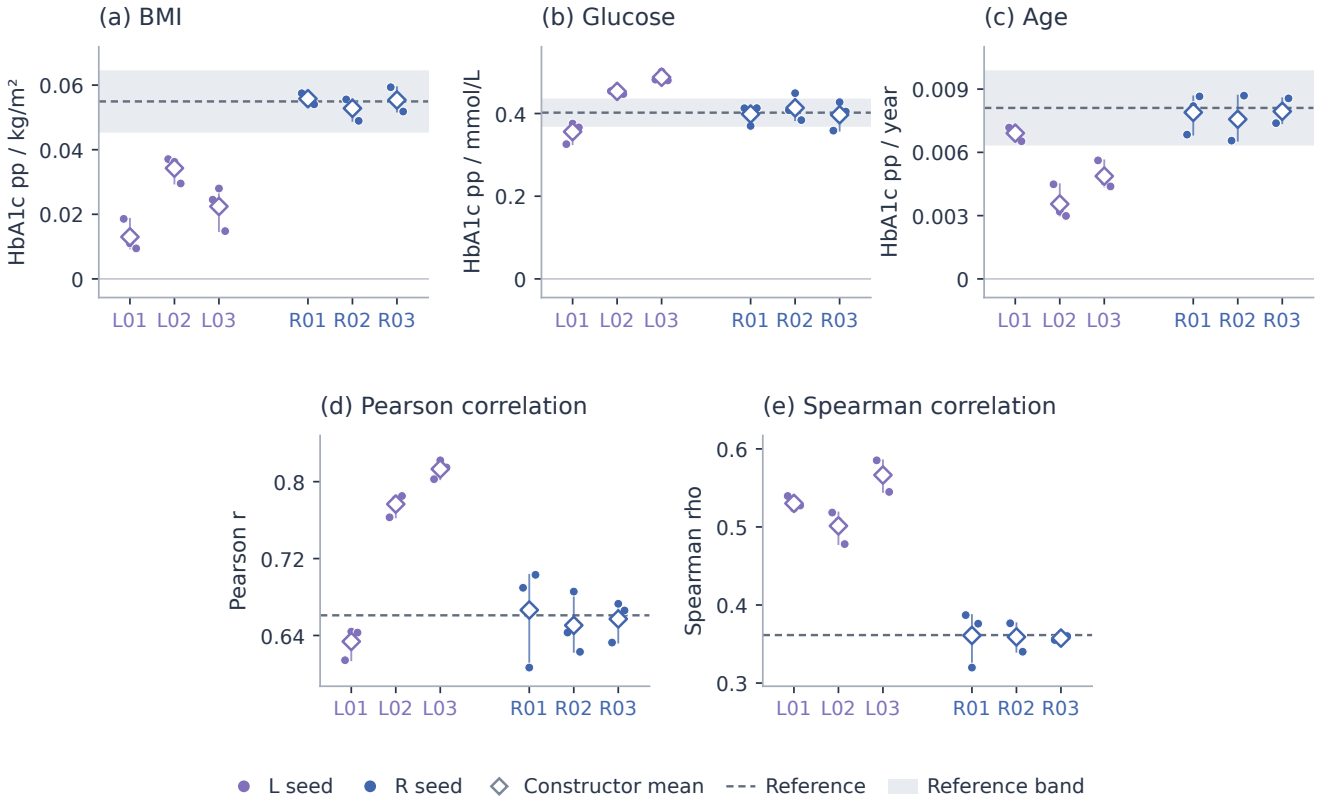


Figure S1: Supplied summaries become jointly analyzable data. Five panels show BMI, glucose and age regression coefficients plus glucose–HbA1c Pearson and Spearman correlations. L01–L03 use literature; R01–R03 additionally receive 30 target summaries and metadata. Points show all three final draws, diamonds constructor means and segments seed ranges. Dashed lines mark reference summaries available to R; regression bands are CHNS community-cluster CR1-t intervals, with no supplied correlation interval. All six constructors, 18 distinct delivery tables, 30 means and 90 seed statistics appear once. Initial and final deliveries were identical after retention; Table S2 gives the complete five-analysis numerical comparison.

Table S2: Joint-statistic reconstruction with available summaries. Ranges span three constructor means over three fixed seeds, not confidence intervals. Regression units are HbA1c percentage points per kg/m², mmol/L and year, respectively.

Analysis	Supplied reference	External literature L	Literature + summaries R
BMI coefficient, age/sex adjusted	0.054946	0.013001–0.034287	0.052821–0.055789
Glucose coefficient, age/sex adjusted	0.402093	0.356064–0.487815	0.396941–0.413715
Age coefficient, sex/BMI/glucose adjusted	0.008107	0.003552–0.006910	0.007569–0.007947
Glucose–HbA1c Pearson correlation	0.660968	0.633764–0.813170	0.650561–0.666438
Glucose–HbA1c Spearman correlation	0.361457	0.501339–0.566477	0.357571–0.360983

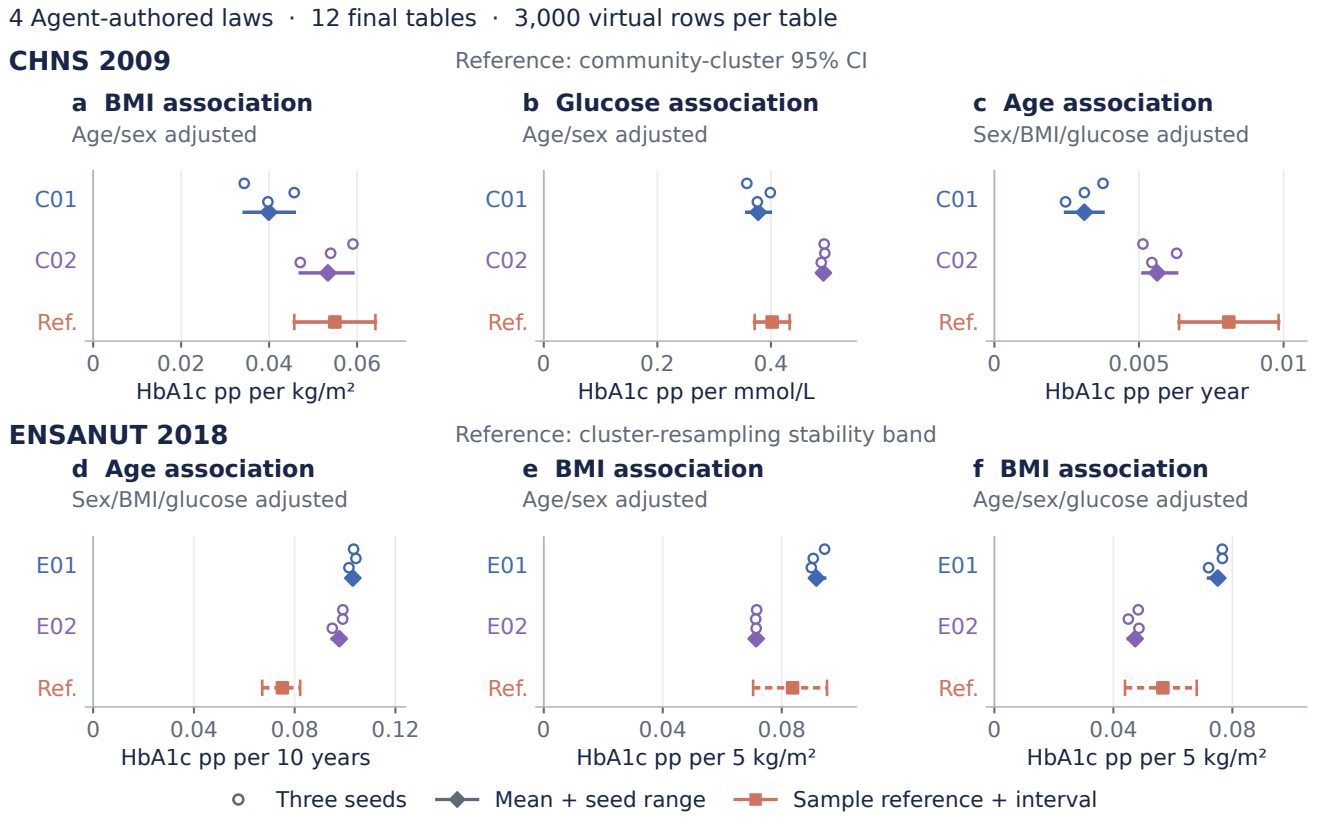


Figure S2: Repeated open-retrieval construction across two biomedical requests. All six prespecified primary analyses are shown for four agent-authored laws and twelve final 3,000-row virtual tables. Blue and purple identify the two constructions within each request. Open circles show seeds 8101–8103, ordered top to bottom within each author row; vertical spacing is solely for legibility. Diamonds and horizontal segments below the circles show the three-seed mean and minimum–maximum range. Coral squares in separate reference rows show previously exposed selected-sample estimates. Solid coral intervals are CHNS community-cluster CR1-t 95% confidence intervals; dashed coral intervals are ENSANUT empirical cluster-resampling stability bands from 399 attempts. These reference intervals have different meanings and neither is an acceptance or equivalence threshold. Axes retain native effect units and zero; pp denotes HbA1c percentage points. All generated coefficients are positive, while both CHNS age estimates are weaker and both ENSANUT age estimates stronger than their references. Seed ranges describe sampling under each fixed law, not confidence intervals or uncertainty over evidence, model choice or transport. The four constructions provide developmental workflow replication on two existing references, not independent clinical replication.

For ENSANUT, E01/E02 means were 0.10308/0.09772 for adjusted age against 0.07513 pp/decade, 0.09165/0.07140 for age/sex-adjusted BMI against 0.08364 pp per 5 kg/m², and 0.07509/0.04733 after additionally adjusting for glucose against 0.05664. Every draw retained the reduction of the BMI coefficient after glucose adjustment, an association pattern rather than causal mediation. Both authors built an age coefficient near 0.10 pp/decade and shared some sources; their agreement does not establish independent evidence convergence.

The complete diagnostics reveal remaining limits. CHNS C01 glucose–HbA1c Pearson correlation was approximately 0.666 versus 0.661, but Spearman correlation was 0.635 versus 0.361. ENSANUT residual HbA1c RMS was approximately 0.295/0.324 versus 0.488, and total SD was 0.362/0.406 versus 0.527. Correct directions and a close Pearson coefficient therefore coexisted with rank and dispersion mismatches. The complete replication report, every diagnostic and Supplementary Table S3 retain the full results. A scalar-type interface defect in the original evaluator was corrected additively on unchanged frozen tables, with its failed output and independent audit preserved; no scientific model was adjusted after comparison.

CHNS BMI/glucose analyses adjust for age and sex; CHNS age adjusts for sex, BMI and glucose. ENSANUT age adjusts for sex, BMI and glucose; its BMI analyses adjust for age/sex, with or without glucose. All Q4 diagnostic values remain in the complete diagnostic table.

The reviewed evidence/specification files distinguish reported statistics, algebraic derivations and assumed relations. The public experiment record includes per-source eligibility, measurement, adjustment, overlap and reporting-quality notes; full third-party originals and participant-like rows remain local.

S4. Longitudinal reconstruction from published artifacts

The HRS recipient-stage example [1] tested a different data product. A fresh agent received published model exports, a variable specification and the intended mixed-model analysis. It authored the recursive conditional generating family with parameter provenance. A separate implementation reconstructed the published law. Each generated 11,509 subjects at 0, 2, 4 and 6 years, producing 46,036 rows per table. No HRS individual records were used, and the source analysis coefficients were construction inputs.

All three agent and three corrected-law tables supported the native R/lme4 model with 16 fixed effects, individual intercepts/slopes and piecewise HbA1c terms around 6.5%, with age centered at 60 and scaled per decade. The below-threshold HbA1c-by-age interaction averaged −0.044635 in agent data, with range −0.051059 to −0.038247, against the supplied −0.046837. The reconstructed law averaged −0.040500. Both preserved its negative direction in all three draws (Figure 2).

Figure 2 shows the estimates and their per-table Wald intervals. The longitudinal analysis record gives all 16 fixed effects and the comparator correction.

The weaker above-threshold interaction was less stable: agent estimates averaged 0.026485 against 0.012867, while corrected-law estimates averaged −0.011691. Overall agent HbA1c mean/SD was 5.8725/0.9676 versus the original published 5.9/1.0. The high-HbA1c subgroup had mean/SD 7.1515/0.5630, compared with original 7.7/1.4 and published synthetic 7.2/0.6. Thus useful source-model behavior coexisted with inadequate upper-group spread.

A CRLF-sensitive parser initially overwrote seven coefficient/SE pairs in our comparator preparation. A separately reviewed source-bound repair produced the three corrected comparator tables without changing the frozen agent outputs. The full record contains nine physical generations, six valid comparison tables and three invalid initial comparator attempts. This was our implementation defect, not a failure of the published method. The incident record and complete effects retain the correction and all outcomes.

S5. Cross-task construction, dispersion and source uses

The next study applied the common workflow to HbA1c analyses in the ENSANUT2018 analytical design and BMI-related TC/HDL analyses in the Turkana study context. Three fresh agent contexts constructed a law for each request, using reviewed external articles, population/measurement metadata and optional literature tools. No participant inputs, target coefficients, target means or prior generators were supplied. The six contexts used the same model service and shared some source evidence. This study therefore tests repeated construction across two requests, with the law chosen by each author, rather than independent clinical replication or a controlled comparison of workflow components.

The HbA1c analyses retained the existing age and BMI estimands and equal allocation to eight sex-by-age cells. The lipid analyses used unweighted age/sex-adjusted BMI coefficients for TC and HDL, plus the Pearson correlation of their residuals after age, sex and BMI adjustment. Lipid eligibility required adults with recognized recorded sex and finite, positive BMI and lipid values. Generated lipid concentrations were converted from mmol/L to mg/dL using the frozen factor 38.67. The reference represents a selected Turkana/Laikipia study sample; fasting, assay transfer and population composition remained qualified in the evidence record. No new participant records were read, downloaded or refitted for this study.

The agents made different scientific choices within the common interface. H01 combined bounded predictors, conditionally truncated glucose and a shifted gamma HbA1c innovation; H02 used bounded beta conditional laws. H03 solved cross-source moment equations under declared dependence and dispersion assumptions, then used conditional beta laws. Their source records included overlapping age, body-size and glycemia evidence [10,17,21,22], with additional check-only/contextual evidence for H01 [14]. The lipid authors combined Gaussian copulas and lognormal margins using geometric summaries from Greiner et al. [25], rural Ashanti rank correlations [26] and arithmetic moments from Agongo et al. [27,28]. Choices about arithmetic versus geometric location, transfer of coefficients of variation, sex strata and age dependence differed among L01–L03. These are applications of established probability families whose numerical use was selected and explained by the agent.

Independent source and implementation reviews checked the statistical meanings, specification/code consistency and finite moments before isolated execution. Each constructor completed one

Table S3: Complete Q4 open-retrieval primary summaries. Means and ranges summarize three final seeds under each fixed construction. Units are HbA1c percentage points per stated predictor increment. No row is omitted; ranges are not confidence intervals.

Request / analysis	Author	Increment	Reference	Generated mean	Seed min–max
CHNS BMI	C01	1 kg/m²	0.054946	0.039923	0.034342–0.045708
CHNS glucose	C01	1 mmol/L	0.402093	0.377538	0.357560–0.398892
CHNS adjusted age	C01	1 year	0.008107	0.003109	0.002461–0.003755
CHNS BMI	C02	1 kg/m²	0.054946	0.053375	0.047088–0.059055
CHNS glucose	C02	1 mmol/L	0.402093	0.492412	0.488649–0.494875
CHNS adjusted age	C02	1 year	0.008107	0.005628	0.005137–0.006301
ENSANUT adjusted age	E01	10 years	0.075132	0.103084	0.101546–0.104315
ENSANUT BMI	E01	5 kg/m²	0.083643	0.091651	0.089940–0.094413
ENSANUT glucose-adjusted BMI	E01	5 kg/m²	0.056638	0.075089	0.072005–0.076661
ENSANUT adjusted age	E02	10 years	0.075132	0.097724	0.094947–0.099135
ENSANUT BMI	E02	5 kg/m²	0.083643	0.071399	0.071232–0.071563
ENSANUT glucose-adjusted BMI	E02	5 kg/m²	0.056638	0.047325	0.045022–0.048592

Table S4: Original-source inventory for Q4 open retrieval. Twelve unique originals supplied sixteen author-specific paper uses. Sharing a source is not independent clinical evidence; source populations are not the target populations.

Original	Reference	Constructing authors
Wu et al., 2017	[16]; PMC5591004	C01
Bao et al., 2010	[7]; PMC2871988	C01
Pani et al., 2008	[10]; PMC2551641	C01, E01, E02
Skogberg et al., 2019	[17]; PMC6458679	C01, E01
Zhang et al., 2016	[18]; PMC5078665	C02
Venkataraman et al., 2012	[19]; PMC3504343	C02
Liu et al., 2016	[20]; PMC5009131	C02
Wisgerhof et al., 2020	[14]; PMC7272077	C02
Huang et al., 2021	[21]; PMC8038122	E01, E02
Ghazanfari et al., 2010	[22]; PMC3075530	E01
Masuch et al., 2019	[23]; PMC6371438	E02
Marini et al., 2017	[24]; PMC5398507	E02

scheduled trial, inspected generated diagnostics and source feedback, and could revise once. All six retained their initial laws. Three fixed final seeds, 9101–9103, produced 18 distinct 3,000-row tables: 54,000 fictional rows from 24 successful physical attempts, including the six trials. No final execution failed and no seed was replaced. Reference comparison began after all six author sessions closed and reused previously exposed aggregate results.

All 54 primary point estimates had the same direction as their corresponding sample estimates, with quantitative agreement varying by analysis (Figure 3; Table S5). HbA1c age effects averaged 0.09954–0.10091 percentage points per decade versus 0.07513 in the reference; age/sex-adjusted BMI effects averaged 0.04892–0.06030 versus 0.08364 per 5 kg/m². With glucose additionally adjusted, H03 averaged 0.05098 versus 0.05664, whereas H01 and H02 averaged 0.03047 and 0.01936. Thus the shared age direction and attenuation after glucose adjustment did not imply recovery of each effect magnitude.

For lipids, mean HDL BMI coefficients ranged from −2.1865 to −1.9868 mg/dL per 5 kg/m² versus −4.6230 in the reference, while TC coefficients ranged from 10.0151 to 11.2609 versus 5.6831. The TC reference sensitivity interval was −3.7535 to 15.1197, so matching its positive point estimate does not establish replication of a definite positive population effect. Residual TC–HDL correlations averaged 0.10828–0.14134 versus 0.51207. The generators preserved the directions of these focal point estimates while substantially underrepresenting the lipid dependence remaining after the specified adjustment.

Dispersion changes the assessment of usefulness (Figure S3). H03's closer glucose-adjusted BMI coefficient accompanied lower

HbA1c marginal SD and full-model residual RMS, 0.3523 and 0.3006 percentage points versus 0.5274 and 0.4885. H02 retained more variation, with corresponding values 0.4825 and 0.4036, but smaller BMI coefficients. L02's TC SD was close to the reference, 43.6778 versus 42.8466 mg/dL, although its residual TC–HDL correlation remained weak. All three lipid laws had HDL SDs below the reference. No construction uniformly dominated across focal effects, dependence and dispersion.

The evaluation retained all 513 prespecified estimates and diagnostics, including the 54 primary outputs. Independent recomputation verified every value and all 324 matched reference comparisons; the maximum numerical difference was 5.33×10^{-14}. The remaining 189 diagnostics lacked matching cached reference estimators and remained generated-only results. Across authors, eight original studies supplied 22 author–paper uses, including context/check-only uses; optional retrieval consumed seven query credits and one original download. All 132 declared source-check entries were unavailable under the implemented semantic rules because weighting or essential operator definitions were unresolved or unsupported (Table S6). Numerical source values still informed reviewed parameter derivations, but this batch provides neither measured numerical source-fit feedback nor a before/after revision benefit. The complete results and reviews retain every endpoint, unavailable comparison and implementation incident, including correction of the independent auditor's identifier parser without changing any generated table.

HbA1c age adjusts for sex, BMI and glucose. HbA1c BMI adjusts for age/sex, with glucose added in the third analysis. Lipid BMI slopes adjust for age/sex; the residual correlation removes age/sex/BMI projections from both lipids on the same rows. H01–H03 belong only to this Q5 HbA1c request, and L01–L03 only to this Q5 lipid request; the L labels do not refer to the earlier Q2 literature arm. All 54 individual primary outputs and all 513 prespecified estimates/diagnostics remain in the complete endpoint table. The 324 matched reference comparisons and 189 generated-only diagnostics retain their original availability labels.

The evidence-use inventory and source/code review retain the source IDs, numerical roles, unresolved definitions and actual retrieval activity. Unresolved weighting and unsupported or incompletely specified operators were not replaced with different statistics. Parameter derivations can therefore use reviewed source information while the corresponding runtime source comparison remains unavailable.

S6. Dependence sensitivity and the supporting construction

The preceding open-search example acquired five seriously reviewed studies [9,10,12,13,14] without bibliographic seeds or desired signs. A fresh constructor selected a sex-stratified Gaussian copula with positive biomarker margins and generated one trial plus three complete 3,000-person tables. It retained the model after eleven approximate and 23 unavailable source comparisons per execution. Secondary BMI coefficients were 0.04217–0.05176 versus 0.05495 in the reused CHNS reference, secondary age was 0.01613–0.01771 versus 0.01530, and primary age was 0.00842–0.01084 versus 0.00811. Primary BMI/glucose were attenuated, and the female point direction and distributional tails remained imperfect. Supplementary Figure S4 retains all seven nonintercept effects and their uncertainty labels.

Before generating data or comparing targets, the source review identified an unreported BMI–glucose dependence. Two valid latent correlations, 0.10 and 0.40, implied conditional BMI coefficients of +0.03422 and −0.01095 while retaining the same margins, other bivariate laws and secondary projection. The delivered model used the declared assumption 0.25. This is an analytical sensitivity, not additional generated cohorts or a causal intervention. It demonstrates why an explicit data-generating law can identify which missing relationship matters to an intended use.

A separate analysis contextualized the same three tables with 100 community-disjoint resplits of the existing CHNS sample, drawing 3,000 people per side and retaining all 36 metrics. Secondary BMI and age discrepancies were on scales comparable with within-sample variation, while primary glucose/female and several joint/tail discrepancies were larger. The complete table preserves the nonexchangeable real-real and generated comparisons. These are descriptive interpretation aids, not a new population or retrospectively selected equivalence cutoff.

S7. Prospective evaluation under the earlier method

In HAALSI, eligible ages started at 40 years and the analysis used the released capillary point-of-care glucose measurement. Fasting was not an eligibility requirement. The fixed BMI and age contrasts were taken from the same additive glucose model with recorded sex. A second model estimated the difference in the BMI slope between women and men, adjusting for age. In Turkana, the two BMI slopes were adjusted for age and recorded sex. The correlation uses the residuals after adjusting both lipid measurements for age, sex and BMI.

Each case had five separately authored agent laws and one prespecified statistical generator (S01). The experiment produced 48 source-context and 36 target-context tables of 5,000 rows each. All authors retained their initial laws after source-only feedback. Initial and final labels therefore refer to the same 36 target tables. The public checkpoint preceding disclosure records work in progress. The final evaluation record documents the complete inventory, one reveal and independent recomputation of 6,704 checks. HAALSI had a prospectively withheld target. Turkana's exact estimates were withheld, with prior investigator exposure to adjacent findings. Later Q5 reused the same 397-person Turkana reference.

The primary rule divided the absolute difference in each slope by the measured outcome's sample SD and used the absolute correlation difference directly. A construction passed when all nine comparisons (three analyses by three seeds) were at most 0.10. The agent counts at thresholds 0.05, 0.10, 0.20 and 0.30 were 1, 3, 5 and 5 of five for HAALSI and 0, 0, 1 and 4 for Turkana. Both S01 laws missed the primary criterion. For HAALSI, the discrepancies for three zero-valued effects were 0.0990, 0.0733 and 0.0770. The frozen decision closed the expansion. The prospective study predates the later interface for documenting numerical evidence use and supplies a separate evaluation of the underlying construction approach.

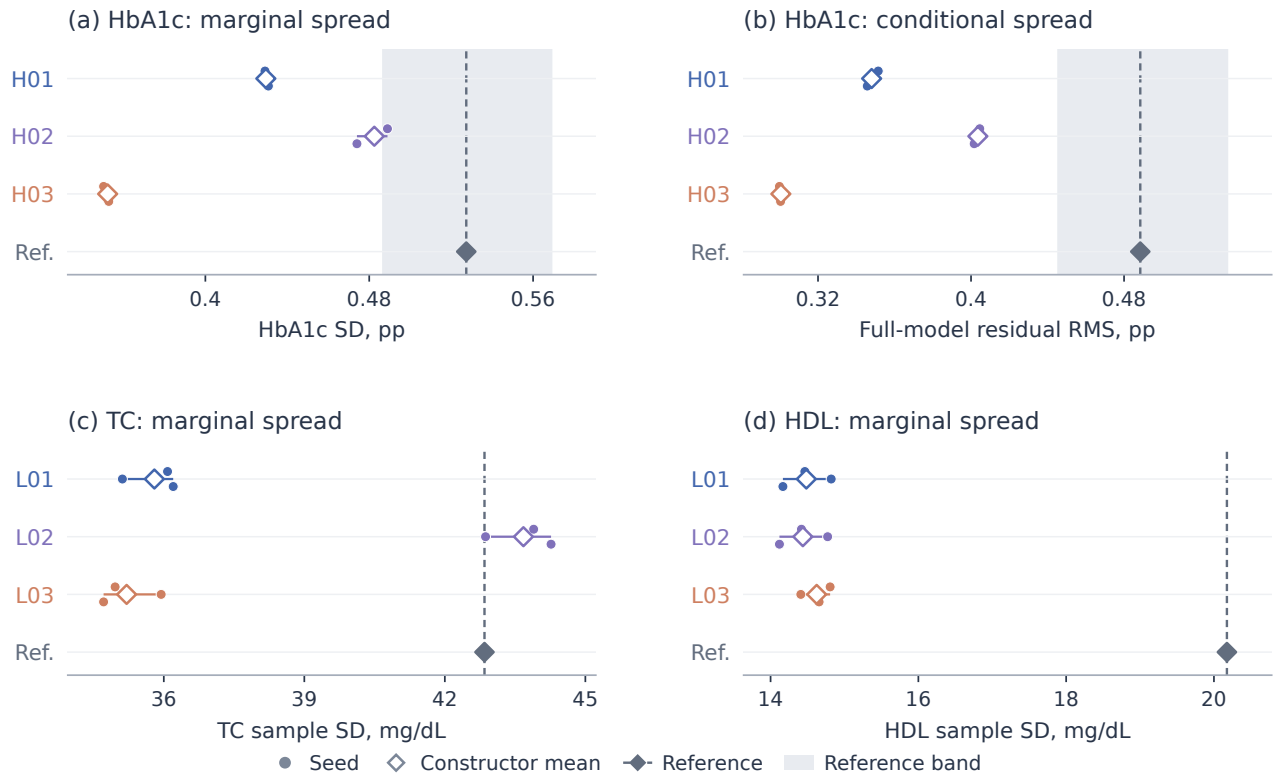


Figure S3: Dispersion across all cross-task constructions. Panels show marginal HbA1c SD, HbA1c full-model residual RMS, and marginal TC and HDL SDs for each construction and its three final draws, retaining all 36 diagnostic values from the same 18 tables. HbA1c uses the population-SD diagnostic; lipids use sample SD. These quantities measure individual or residual spread, not uncertainty in a regression coefficient. H03's relatively close adjusted BMI effect in Figure 3 coexists with reduced HbA1c spread, and L02's close TC SD coexists with weak adjusted lipid dependence. Seed ranges reflect Monte Carlo variation conditional on the selected law. Gray HbA1c bands are empirical cluster-resampling stability bands; no reference interval is supplied for lipid SDs.

S8. Full records and retained implementation incidents

The supplementary results preserve the study designs, estimators, source inventories, implementation incidents and numerical audits.

The main study, summary-assisted study, statistical supplement, HRS example, preceding open-search construction, unified workflow replication and cross-task evidence-use study preserve separate frozen identities and complete endpoint outputs. The cross-task protocol, complete endpoint table, independent numerical review and scientific interpretation document the new unit. Earlier live A/B results, including the conditioned BMI mismatch and source-replay unavailability, remain in the extended record. Figure updates and added tables consume reviewed aggregates only; manuscript integration creates no cohorts or clinical fits. In the main study, seven authors inspected exploratory outputs and kept their laws, and thirteen finalized without exploration. The two earlier live authors both kept their models after the available review.

Individual real/generated tables, third-party originals and credentials remain local; clinical reruns require lawful access and the appropriate environment. Historical model identifiers and token costs are incompletely metered.

The historical supplement retains earlier enrichment and direct-row construction, the divergent-moment repair, failed transport criteria, tail defects and revision trade-offs. The v0.16 reading draft preserves the earlier layout.

Table S5: Complete Q5 cross-task primary summaries. Each row summarizes three fixed final draws from one law. All 18 author–analysis combinations are retained. HbA1c entries are percentage points (pp); lipid slopes are mg/dL per 5 kg/m². Ranges are seed minima and maxima, not confidence intervals or evidence/model uncertainty.

Request / analysis	Author	Units / increment	Reference	Generated mean	Seed min–max
HbA1c adjusted age	H01	pp / 10 years	0.075132	0.099545	0.094075–0.103606
HbA1c BMI	H01	pp / 5 kg/m²	0.083643	0.060295	0.055769–0.063938
HbA1c glucose-adjusted BMI	H01	pp / 5 kg/m²	0.056638	0.030469	0.027956–0.034291
HbA1c adjusted age	H02	pp / 10 years	0.075132	0.099539	0.098162–0.100396
HbA1c BMI	H02	pp / 5 kg/m²	0.083643	0.048923	0.029090–0.065499
HbA1c glucose-adjusted BMI	H02	pp / 5 kg/m²	0.056638	0.019363	0.002308–0.030777
HbA1c adjusted age	H03	pp / 10 years	0.075132	0.100912	0.096356–0.105015
HbA1c BMI	H03	pp / 5 kg/m²	0.083643	0.056649	0.048449–0.063758
HbA1c glucose-adjusted BMI	H03	pp / 5 kg/m²	0.056638	0.050978	0.044994–0.056901
HDL BMI	L01	mg/dL / 5 kg/m²	−4.623004	−2.186526	−2.286805–−2.118624
TC BMI	L01	mg/dL / 5 kg/m²	5.683108	10.393977	9.992186–10.676486
TC–HDL residual correlation	L01	dimensionless	0.512070	0.111234	0.108517–0.113140
HDL BMI	L02	mg/dL / 5 kg/m²	−4.623004	−2.100390	−2.194806–−2.035712
TC BMI	L02	mg/dL / 5 kg/m²	5.683108	11.260864	10.811415–11.565687
TC–HDL residual correlation	L02	dimensionless	0.512070	0.108283	0.105395–0.110194
HDL BMI	L03	mg/dL / 5 kg/m²	−4.623004	−1.986781	−2.122835–−1.913541
TC BMI	L03	mg/dL / 5 kg/m²	5.683108	10.015112	8.617345–11.698752
TC–HDL residual correlation	L03	dimensionless	0.512070	0.141343	0.121087–0.155368

Table S6: Q5 original-source uses and unavailable feedback. Eight distinct originals supplied 22 author–paper uses, including context/check-only evidence. All six authors retained their initial laws. The 132 source-check entries were unavailable, not successful numerical comparisons or failed generators. The inventory counts the three lipid originals, with the Agongo correction [28] attached to its source rather than counted as a separate original study.

Author	Original-source references	Paper uses	Unavailable source checks
H01	Pani [10]; Wisgerhof [14]; Skogberg [17]; Huang [21]; Ghazanfari [22]	5	13
H02	Pani [10]; Skogberg [17]; Huang [21]; Ghazanfari [22]	4	18
H03	Pani [10]; Skogberg [17]; Huang [21]; Ghazanfari [22]	4	32
L01	Greiner [25]; Obirikorang [26]; Agongo [27,28]	3	29
L02	Greiner [25]; Obirikorang [26]; Agongo [27,28]	3	19
L03	Greiner [25]; Obirikorang [26]; Agongo [27,28]	3	21
Total	8 distinct original studies	22	132

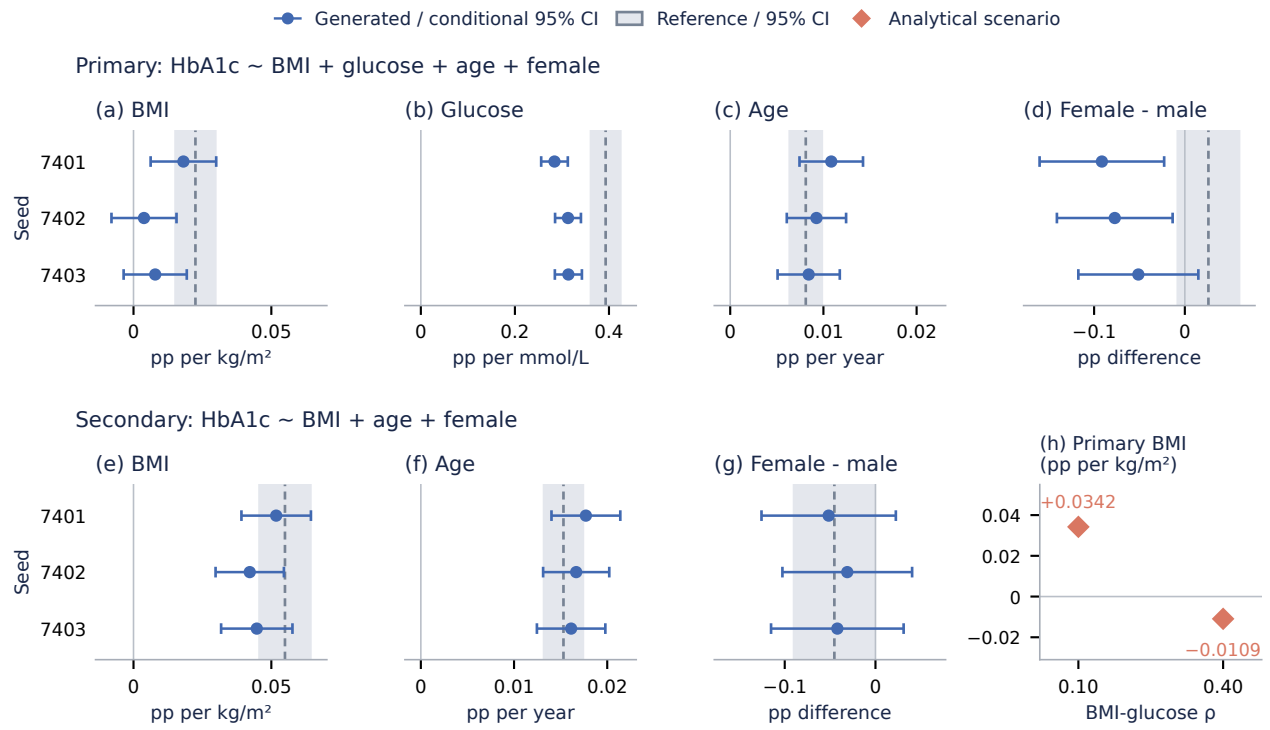


Figure S4: Analysis-specific usefulness and an unresolved joint dependence. Panels (a–d) show all primary-model nonintercept effects and (e–g) all secondary-model effects. Blue points are three fixed 3,000-person draws with conditional 95% HC3-t intervals. Gray dashed lines and bands are exposed CHNS reference estimates and 95% community-cluster CR1-t intervals; solid lines mark zero. All effects retain native units (pp: HbA1c percentage points), with shared limits for matching predictors. Generated intervals omit source, structural and transport uncertainty; overlap is not equivalence. Panel (h) displays two pre-generation analytical scenarios: latent BMI–glucose correlations 0.10 and 0.40 imply conditional BMI coefficients +0.03422 and −0.01095 while leaving marginal laws, other bivariate laws and the secondary projection unchanged. The delivered model used 0.25. These are not new draws, a fitted curve or a causal intervention. Retained initial/final stages alias the same three tables.

Table S7: Six prespecified analyses under the earlier workflow. The columns report the measured-sample estimate, the minimum and maximum across 15 agent draws, the number agreeing in point sign, and the range of the three draws of the prescribed statistical law. The ranges describe draws under the candidate models. HAALSI glucose contrasts are in mmol/L, lipid slopes in mg/dL and the adjusted correlation is unitless.

Analysis	Measured	Agent range	Same sign	S01 range
HAALSI glucose, BMI per 5 kg/m²	0.3138	0.3285 to 0.5698	15/15	−0.0174 to 0.0180
HAALSI glucose, age per 10 years	0.2322	0.1136 to 0.3005	15/15	−0.0170 to 0.0077
HAALSI female-minus-male BMI slope per 5 kg/m²	−0.2440	−0.2487 to 0.1903	8/15	−0.0524 to 0.0046
Turkana HDL, BMI per 5 kg/m²	−4.6230	−3.5509 to −1.8173	15/15	−0.1120 to 0.0710
Turkana TC, BMI per 5 kg/m²	5.6831	10.2822 to 15.0413	15/15	−0.4657 to 0.0334
Turkana residual TC–HDL correlation	0.5121	0.1162 to 0.5361	15/15	0.3625 to 0.3840

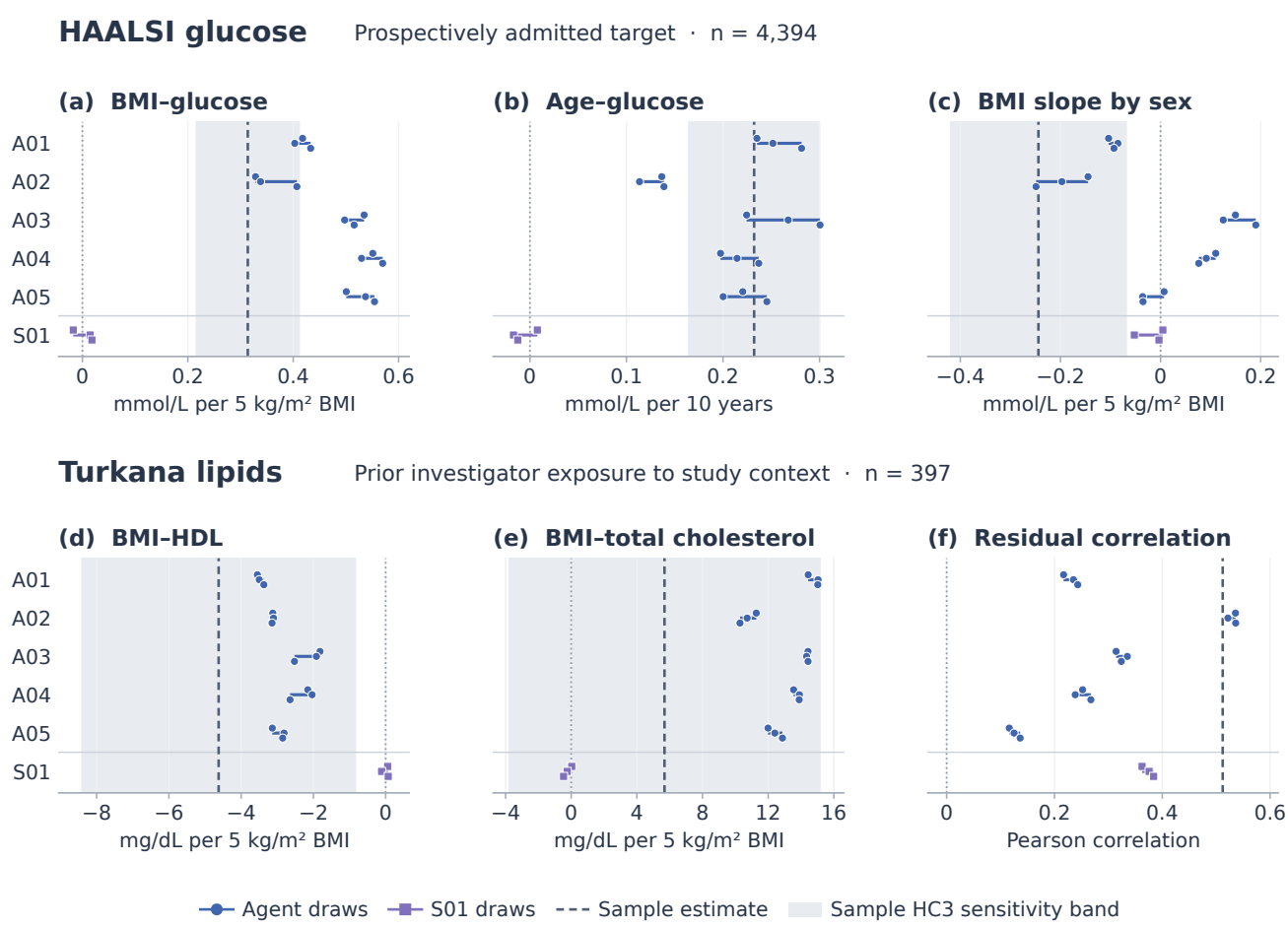


Figure S5: Earlier prospective evaluation of whole-cohort generation. Blue circles show the three draws of each agent law and purple squares the three draws of the prescribed statistical law (S01). Short segments span draws. Dashed lines mark measured-sample estimates, and shaded bands give approximate 95% HC3 sensitivity intervals for the five slopes. The residual correlation has no reference interval. HAALSI glucose main effects were directionally stable across agent draws. The sex difference in the BMI slope varied by construction. Turkana's total cholesterol point estimate had a reference interval that included zero. Its quantitative estimates were withheld before generation despite prior exposure to related published findings.